

SUDARSHANA KARKALA

Director of Engineering (AI + Battery + Energy + eMobility)

Open to Architect / CTO / Founding Engineer roles in AI + eMobility + Energy platforms

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SUMMARY

- Director / Principal Architect with 20+ years of experience building AI-driven mobility, energy, and IoT platforms.
- Expert in GenAI, Agentic AI, EV Battery Systems, Telematics, and Mobile + Cloud architectures.
- Bridging AI with physical systems (EV batteries, mobility, IoT) to enable intelligent, real-time, safety-critical decision systems
- Currently focused on **AI-powered EV Battery Safety, Predictive Diagnostics, and Intelligent eMobility Platforms**.
- Strong track record of delivering **scalable, secure, production-grade systems used globally**.

KEY ACHIEVEMENTS

- Built mobile platforms used by **3M+ users** with **4.7★ App Store rating (McAfee)**
- Led **30+ engineers** delivering global solar, EVSE, and IoT platforms at **Enphase Solar Energy**
- Architected **AI-powered EV Battery Fire Prevention System** (predictive safety platform)
- Designed **Battery Digital Twin & Telematics platforms** for **Autonomous EV** ecosystem
- Delivered **secure mobile + IoT platforms** with BLE, cloud, and real-time telemetry
- Trained **200+ engineers** in EV systems, GenAI, and mobility architectures

CORE EXPERTISE

- **GenAI & Agentic AI:** OpenAI/GPT, Gemini, Claude APIs, LLMs, RAG, Prompt Engineering, LangChain (conceptual), Multi-Agent Systems, Scikit-learn, TensorFlow
- **EV & Battery Systems:** BMS, Diagnostics, Thermal Safety, Digital Twin
- **Telematics & IoT:** BLE, CAN, MQTT, Device Provisioning
- **Mobile Architecture:** iOS (SwiftUI), Android, MVVM, Clean Architecture
- **Cloud & Data:** Firebase, AWS, Event-driven systems
- **Security:** Encryption, Secure Device Communication, Automotive Cybersecurity

CURRENT ROLE

Thasmai Infotech Private Limited | Bengaluru | Nov 2025 – Present

Co-Founder, Director of Engineering – GenAI & EV Battery Intelligence | Portfolio: <https://battery.ev.engineer>

Designed AI-driven EV Battery Safety Platform to predict and prevent thermal runaway and fire risks in EV batteries using real-time data, simulation models, and intelligent decision agents.

Project: AI-Powered EV Battery Fire Prevention System (GenAI + Agentic AI Battery Safety Platform)

- Architected **Agentic AI system** with Monitoring, Risk, and Alert agents for real-time battery safety decisions
- Built **AI/ML models for thermal runaway prediction**, improving early fault detection using multi-sensor data
- Designed **GenAI-powered diagnostic assistant** for technicians (battery insights, failure explanations)
- Implemented **RAG-based pipeline** combining NASA datasets, PyBaMM simulations, and real-time telemetry
- Developed **battery digital twin simulation layer** for predictive analytics and safety modeling
- Designed **event-driven AI architecture**: Data → Prediction → Risk Score → Action
- Enabled **predictive maintenance and proactive safety alerts** for EV users and fleets
- Improved early fault detection accuracy using AI-based models on simulated and real-world datasets
- **Tech Stack:** Swift, SwiftUI, Kotlin, BLE, MQTT, Firebase, Python, LLM (RAG), AES, CRC16, AWS, GCP

Principal Architect / Consultant – eMobility, GenAI & Agentic AI | Bengaluru | Dec 2024 – Nov 2025

Project: Smart Two-Wheeler Telematics & Battery Platform (Connected EV + GenAI + IoT)

- Designed **end-to-end telematics architecture** (Mobile + BLE + Cloud + Device layer)
- Built **secure BLE communication system** with AES encryption, CRC validation, and device authentication
- Implemented **QR-based digital identity system** (Device Mapping → Digital Twin representation)
- Developed **Agentic AI workflows** for fleet monitoring, battery optimisation, and decision automation
- Integrated **GenAI assistant for fleet insights and operations** (natural language queries)
- Designed **offline-first lock/unlock system**, enabling operation without network connectivity
- Built **vendor-agnostic modular architecture** supporting multiple lock and EV device types
- Enabled **battery swapping intelligence layer** using usage and telemetry analytics

Project: Architected Battery Pack Aadhaar System for lifecycle tracking and compliance

- Designed **Battery Digital Twin** using AI-driven diagnostic pipelines
- Built **GenAI + Agentic AI architectures** for predictive maintenance and fault analysis
- Developed **SDV and telematics platforms** (diagnostics, telemetry, cybersecurity)
- Delivered mobile applications for EV control, tracking, and diagnostics

Client Engagements: Delivered consulting and architecture solutions for automotive, IoT, and AI-driven engineering platforms across multiple enterprise clients

- **Schaeffler India:** GenAI for Automotive & Embedded Engineering workflows
- **Crompton Greaves Consumer Electricals, Mumbai:** Smart Home IoT – iOS Architecture & BLE-based systems
- **iTelematics Software Private Limited** - eMobility, GenAI & Agentic AI
- **Skill-Lync:** Trainer & Consultant – iOS, GenAI, Agentic AI
- **iFab innovations** - Automotive & **EV Motor predictive maintenance** using **agentic AI workflows**

PREVIOUS WORK EXPERIENCE

Enphase Solar Energy Private Limited | Bengaluru | Jul 2021 – Dec 2024

Sr. Engineering Manager – Mobile Apps (iOS, Android, EVSE, IoT, Cloud)

Led architecture and engineering of **scalable mobile** and **IoT platforms** for Enphase's solar, EVSE, and energy storage ecosystem, enabling global device commissioning and digital transformation.

Key Responsibilities

- Led **30+ engineers** across iOS, Android, IoT, and EVSE domains
- Owned architecture of **Enphase Installer Toolkit (ITK)** for global device commissioning
- Designed **device provisioning workflows** (Wi-Fi, BLE, Gateway connectivity)
- Built secure mobile + IoT communication architecture
- Delivered apps with **4.3★+ rating** and strong global adoption
- Modernised architecture using **MVVM + Clean Architecture**
- Improved system reliability and reduced field failures
- Introduced GenAI-based chatbot assistant for technician support and troubleshooting
- Reduced commissioning failures by 25%

McAfee Software (India) Private Limited | Bengaluru | Oct 2011 – Jul 2021

Engineering Manager – Mobile Security & Telematics

Led architecture and development of **McAfee Mobile Security (iOS)** and **Telematics Security applications**, delivering large-scale consumer security features, IoT integrations, and data-driven insights.

Key Contributions & Achievements

- Architected and built McAfee Mobile Security (iOS) from scratch
- Delivered 3M+ downloads, 4.7★ rating (150K+ reviews)
- Developed VPN, Wi-Fi Security, Safe Browsing, Identity Protection features
- Built Telematics Security App using OBD-II (driving behavior analytics)
- Designed encryption, secure storage, and privacy-first architecture
- Implemented analytics systems (AppsFlyer, GA, Power BI)
- Acted as Scrum Product Owner (CSPO) – roadmap, backlog, product strategy
- Led security, privacy, and compliance initiatives

eNovate Media Solution Private Limited | Bangalore, India | Dec 2010 – Aug 2011

Technical Leader – iOS

- Led a team of 6+ iOS developers, driving design and delivery of mobile applications.
- Developed multiple educational **eReader** applications for **Pearson**, focusing on usability and performance.
- Mentored team on UML design, architecture principles, and iOS development best practices.
- Ensured high-quality delivery through code reviews, design discussions, and team guidance.

Ness Technologies Private Limited | Bangalore, India | Jul 2009 – Nov 2010

Senior Software Engineer – iOS

- Developed travel and airline mobile applications for enterprise clients including Amadeus.
- Contributed to CheckMyTrip Mobile Companion and MERCI Airline Mobile App (4.5★ App Store rating).
- Designed and implemented key UI modules: Trip List, Trip Details, Help/About Chronological sorting of trip segments
- Implemented core functionalities: XML parsing, data handling, Localisation and multi-language support, Native iOS integrations
- Designed “PlanMyTrip” prototype, a cross-platform travel planning app (iPhone, Blackberry, J2ME).

NMS Communications Private Limited | Bangalore, India | Jun 2005 – Jul 2009

Senior Software Engineer – Mobile Platforms

- Developed mobile applications across J2ME, Windows CE, and Symbian OS platforms.
- Contributed to Full Track Music platform (LiveWire Mobile):
 - Authentication, billing, and provisioning workflows, Integration using Java, XML, and Web Services
- Built features for MyCaller (Ringback Tone / Hello Tune):
 - GUI development, SMS alerts, XML encoding, Media download and contact (PIM) access
- Developed Media Messaging client (prototype) for multimedia communication.
- Built Peer-to-Peer media sharing application (J2ME prototype).
- Contributed to IMS client features: Meeting request and voting systems, Technologies: C, C++, SIP, MSRP

Passionate about building AI systems that interact with real-world physical environments, improving safety, efficiency, and intelligence in next-generation mobility and energy platforms

EDUCATION & CERTIFICATIONS

- **BE – Information Technology**, National Institute of Technology Karnataka (NITK), Surathkal, 2001-2005
- **Electric Vehicle Engineering & Development**, IIT Madras (CODE), 2024 - 2026
- **Certified Scrum Product Owner (CSPO)**, **Generative AI**, Google Cloud Skills Boost
- **Certified EV Technology Officer (EVTO™)**, EV Society™ - **Ongoing** - Due Date - Mar 2027

TRAINING & INDUSTRY CONTRIBUTION

- Trained **2000+ engineers** on EV systems, GenAI, Agentic AI, mobile & cloud architecture
- Delivered corporate and academic programs on **EV Battery Software** and **Automotive AI**